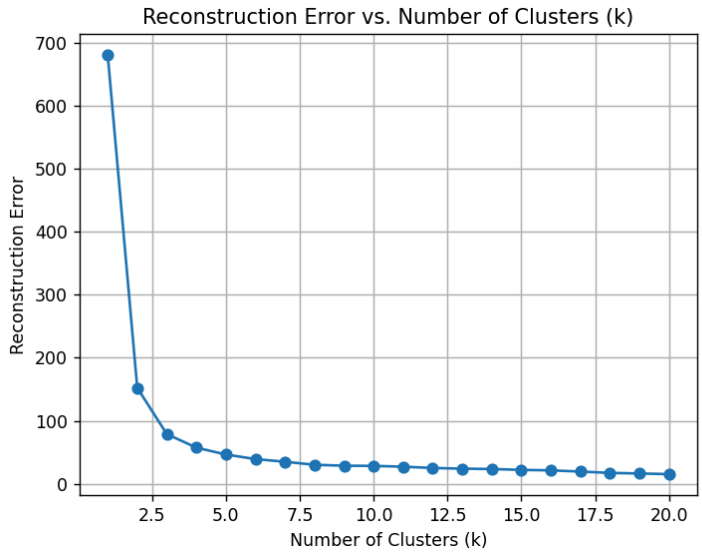
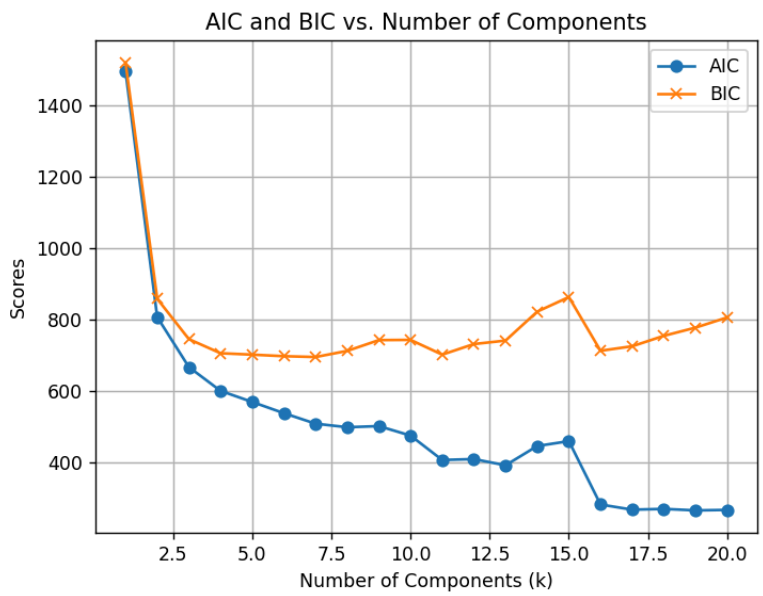
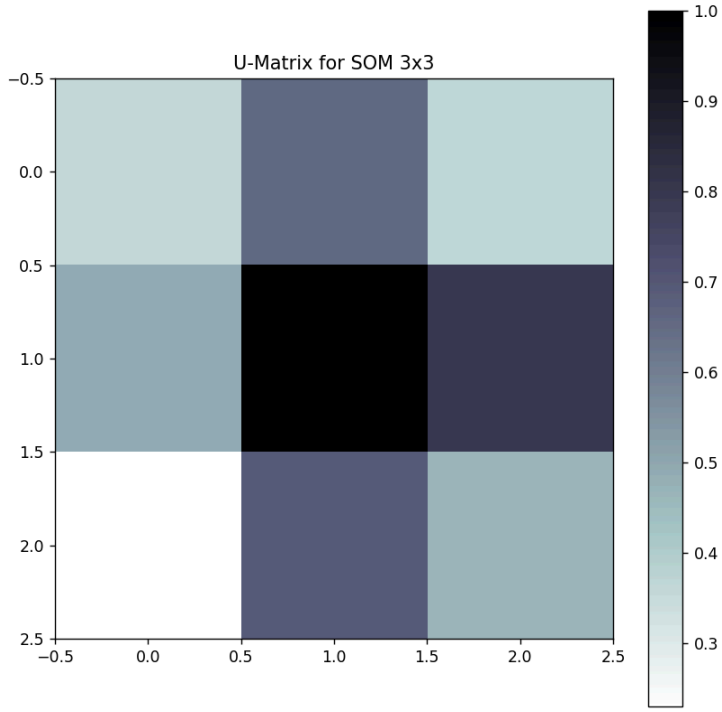
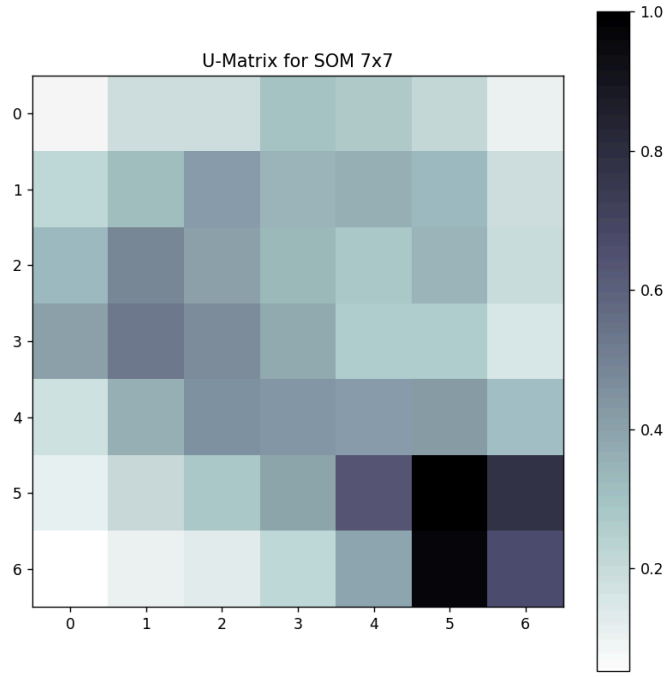
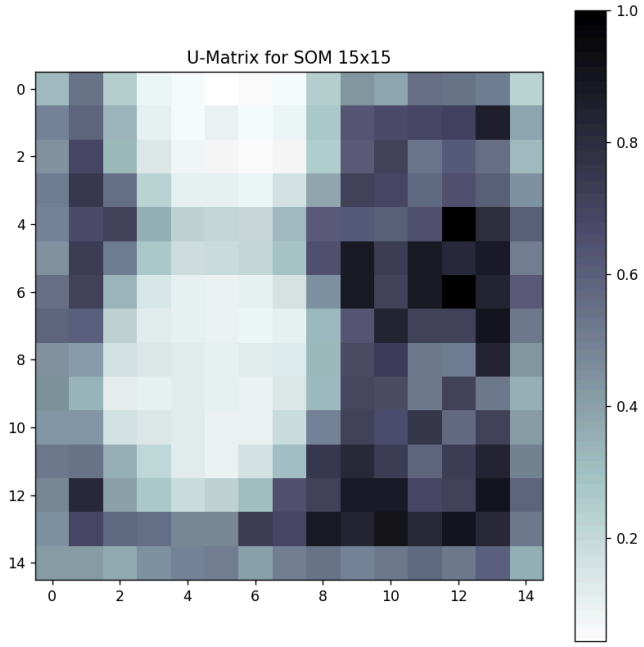
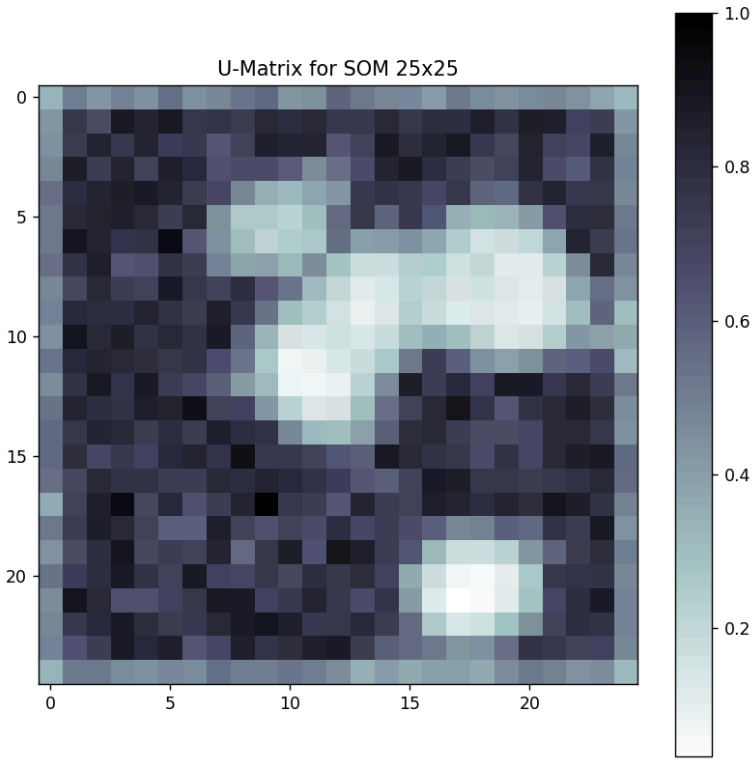

Reconstruction Error vs. Number of Clusters (k)
Reconstruction Error
Number of Clusters (k)
0
100
200
300
400
500
600
700
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0

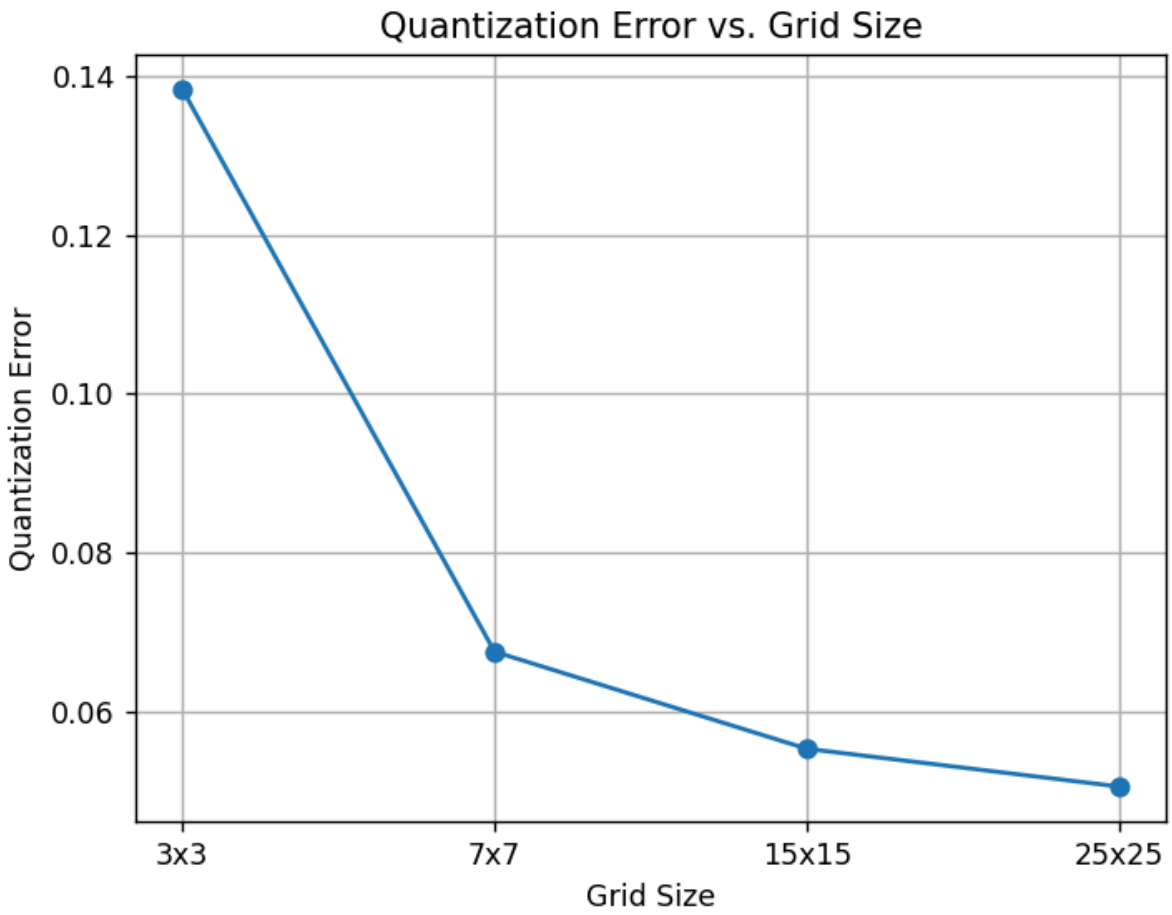

AIC and BIC vs. Number of Components
AIC
BIC
Scores
Number of Components (k)
400
600
800
1000
1200
1400
2.5
5.0
7.5
10.0
12.5
15.0
17.5
20.0


U-Matrix for SOM 3x3
−0.5
0.0
0.5
1.0
1.5
2.0
2.5
1.0
0.9
0.8
0.7
0.6
0.5
0.4
0.3


U-Matrix for SOM 7x7
0
1
2
3
4
5
6
1.0
0.8
0.6
0.4
0.2


U-Matrix for SOM 15x15
0
2
4
6
8
10
12
14
1.0
0.8
0.6
0.4
0.2


U-Matrix for SOM 25x25
0
5
10
15
20
1.0
0.8
0.6
0.4
0.2


Quantization Error vs. Grid Size
Quantization Error
0.14
0.12
0.10
0.08
0.06
3x3
7x7
15x15
25x25
Grid Size

```
Part 1: k-Means Clustering
Elbow found at k=3
Confusion Matrix for k=elbow_k:
[[ 0 50  0]
 [ 3  0 47]
 [36  0 14]]
Accuracy for k=elbow_k: 0.8866666666666667
Confusion Matrix for k=3:
[[ 0 50  0]
 [ 3  0 47]
 [36  0 14]]
Accuracy for k=3: 0.8866666666666667

Part 2: Gaussian Mixture Models (GMM)
AIC elbow at k=3
BIC elbow at k=3
Confusion Matrix for k=aic_elbow_k:
[[ 0 50  0]
 [ 1  0 49]
 [36  0 14]]
Accuracy for k=aic_elbow_k: 0.9

Part 3: Self Organizing Map (SOM)
Quantization error for grid size (3, 3): 0.1383
```